

ROVER LAB, DOSSIER (TOI-700 d mission)

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Planet

Name: TOI-700 d

Planet type: Earth-sized rocky planet in the habitable zone of a red dwarf

Surface temperature: ≈ -4 °C equilibrium estimate; likely tidally locked

Surface pressure: Not measured; no atmosphere detected with current telescopes

Gravity: ≈ 1.2 – 1.7 g (the planet's mass is not yet precisely settled, present as an estimate)

Atmosphere: Unknown/undetected. Published climate scenarios are theoretical models, not measurements.

Mission Briefing

TOI-700 d is an Earth-sized rocky world orbiting safely within the habitable zone of a small, quiet red dwarf. Likely tidally locked, it may have a permanent dayside and nightside separated by a band of eternal twilight. The rover 'Meridian-1' has been sent to a reddish, iron-stained basin in this twilight band, its colouring reminiscent of the hematite-rich terrain found by rovers exploring ancient dried lakebeds on Mars. Orbital spectrometry suggests the basin may hold a chloride brine rich enough in dissolved iron to be worth processing into iron oxide for the station's shielding, but only a ground sample can confirm it.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Reddish, hematite-like basin in the twilight band

Description: A shallow basin with a distinctive rusty-red crust, cut by cracked, sun-baked mineral ridges.

Identity: Iron(III) chloride solution (real sample, suggested ≈ 0.2 M), medium concentration. REAL SAMPLE GROUP.

Known Properties

Colour: Amber-orange, moderate intensity

Odour: Faint, sharp

Solubility: Fully miscible with water

Experiment Results

Measure pH: Acidic, pH around 1.8–2.1

Measure density: 1.15 g/mL

Test electrical conductivity: Moderately high

Add a magnet: No attraction

Add potassium thiocyanate solution: Turns deep blood-red almost immediately

Add sodium hydroxide solution: Forms a moderate reddish-brown gelatinous precipitate

Heat a small sample gently: Slight darkening

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

REAL SAMPLE GROUP, assign real FeCl_3 , medium concentration (suggested $\approx 0.2 \text{ M}$). The 'hematite-like basin' framing deliberately echoes a real Mars connection: NASA's Opportunity rover found iron-oxide-rich 'blueberries' (hematite spherules) in ancient, evaporated lake sediments at Meridian Planum, a genuine real-world precedent worth telling students directly.

Real-science notes: TOI-700 d is a genuine, well-established Earth-sized habitable-zone planet, ~ 101 light-years away, discovered by TESS in 2020 (the first Earth-sized habitable-zone planet TESS found), orbiting its red dwarf star every 37.4 days. Its mass is still not precisely settled: an early statistical estimate gave $\sim 1.7\text{--}1.9$ Earth masses, while a newer (2024–25, still preliminary) radial-velocity measurement suggests ~ 2.4 Earth masses, a good example to show students that even 'known' exoplanet properties keep being refined by better data. No atmosphere has been detected; all climate/'twilight band' scenarios in the literature (e.g. Gilbert et al. 2020) are computer models of hypothetical atmospheres, not observations, make this distinction clear if students research this planet for the conference.

Sources: NASA (svs.gsfc.nasa.gov/13496); Gilbert et al. 2020 (arxiv.org/pdf/2001.00955).

Cross-subject link: Biology selective pressure for this planet, possible tidal locking means a permanent dayside and nightside with extreme temperature contrast across the twilight boundary.